

EE2 Circuits and Systems

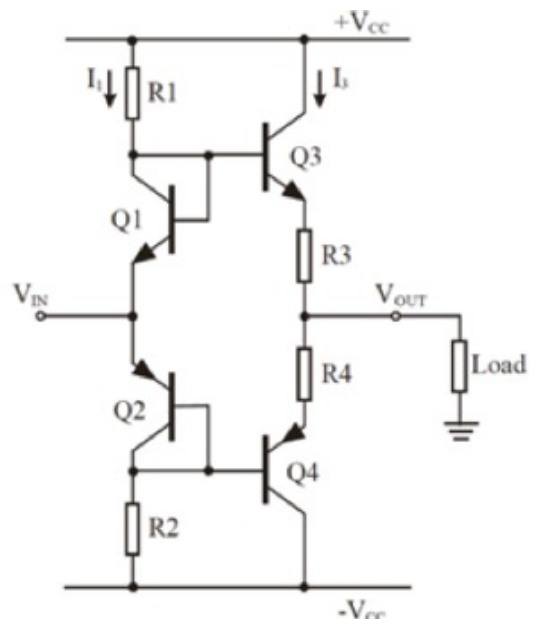
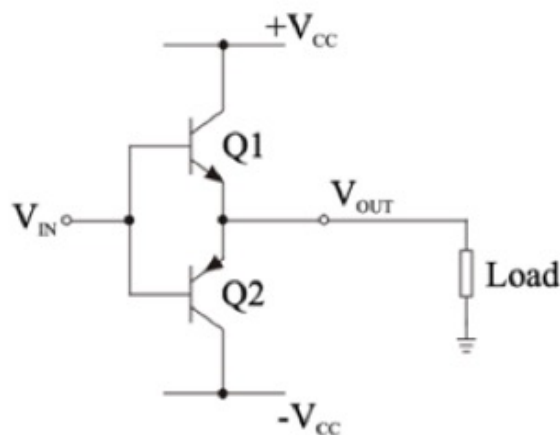
Problem Class 4 (Problem Sheet 2)

Prof Peter Y. K. Cheung
Imperial College London

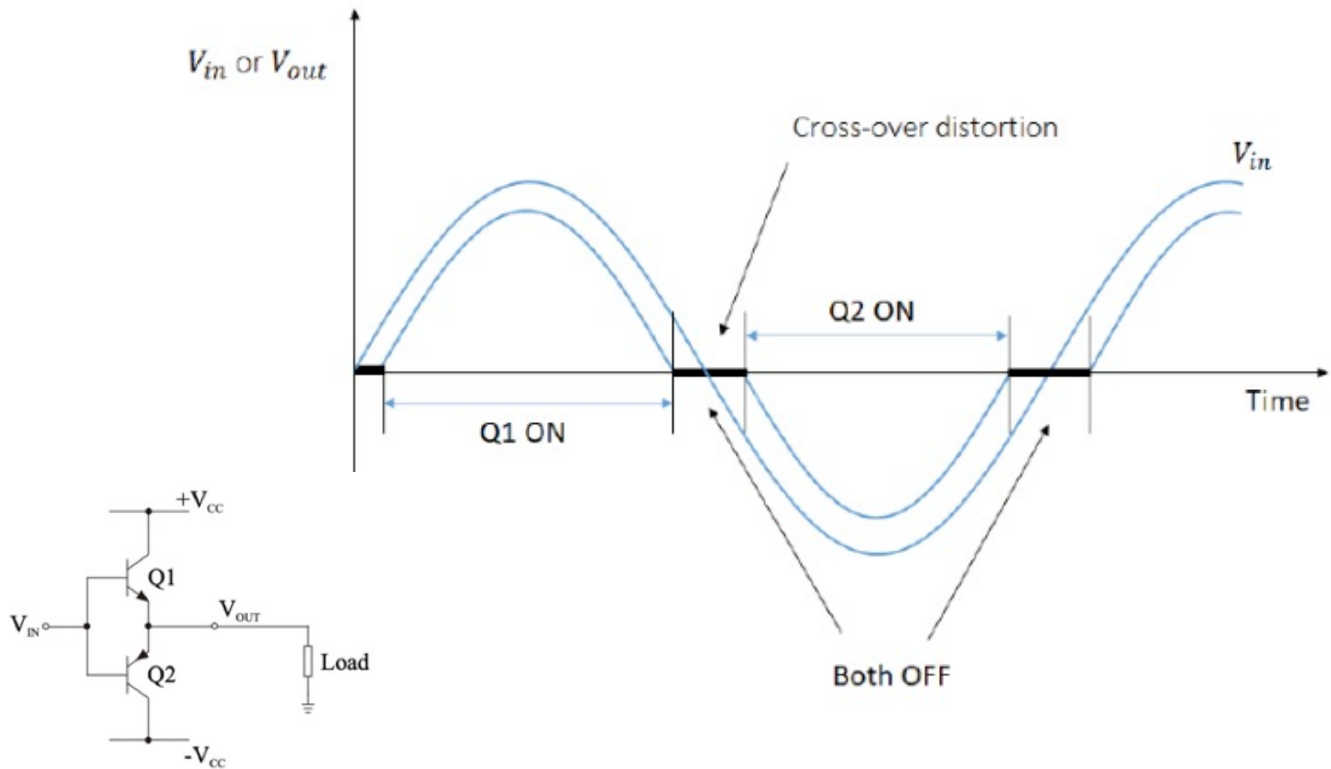
URL: www.ee.ic.ac.uk/pcheung/teaching/EE2_CAS/
E-mail: p.cheung@imperial.ac.uk

Test Yourself - Problem 1 (Sheet 2 Q1)

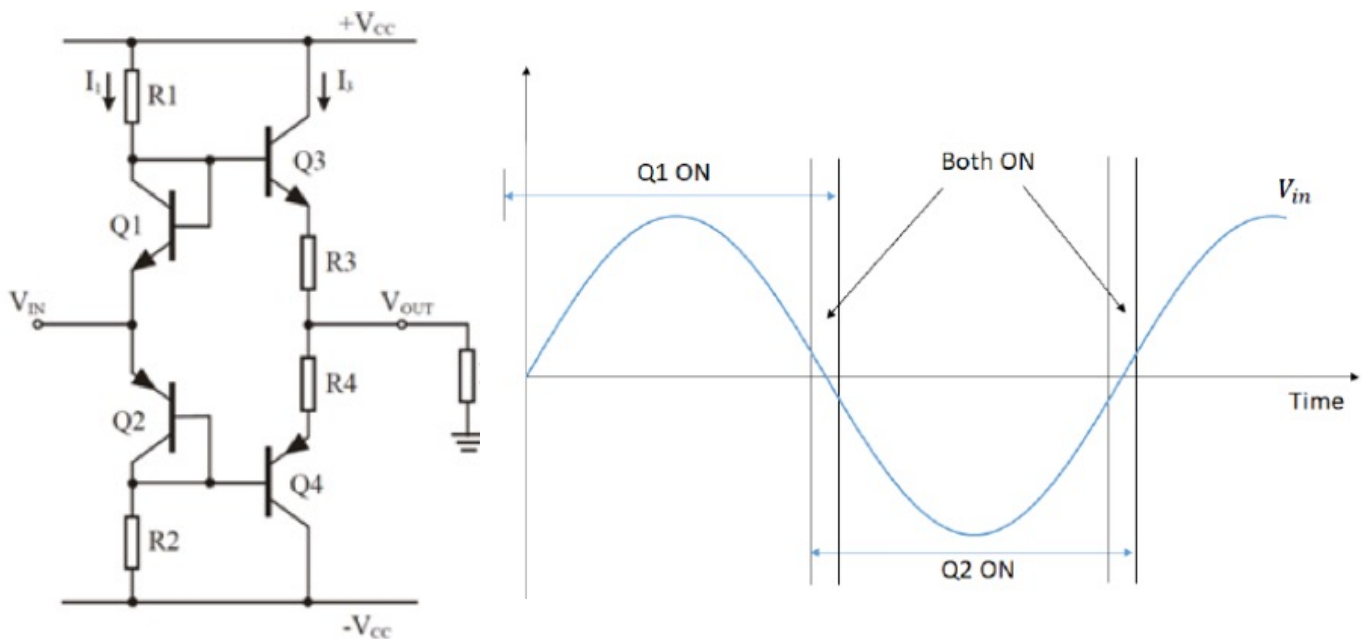
Shown in *Figure Q1a* is an output circuit that drives a resistive load. If V_{IN} is a sinewave, plot the waveform for V_{OUT} . The output driver circuit is now modified to that shown in *Figure Q1b*. Explain how this circuit works and plot the waveform for V_{OUT} for a sinusoidal input.



Test Yourself - Solution 1 (Sheet 2 Q1)

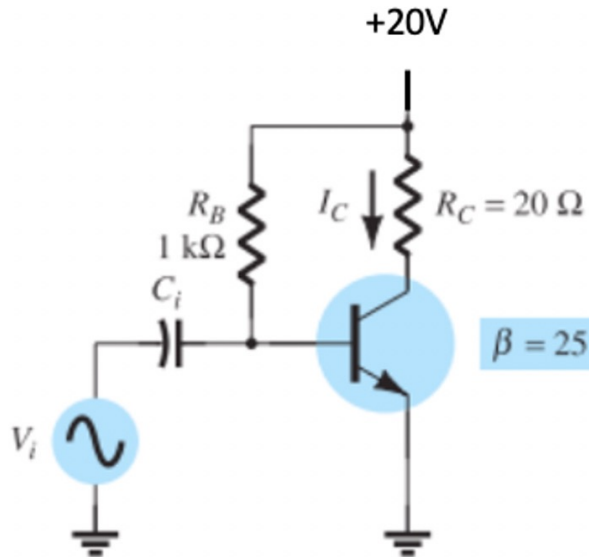


Test Yourself - Solution 1 (Sheet 2 Q1)

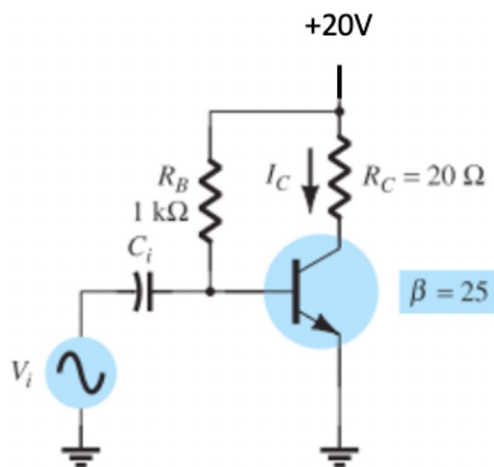


Explain it - Problem 2 (Sheet 2 Q2)

Figure Q2 shows a class A amplifier circuit with $V_{CC} = 20V$. R_C is the load resistance. Assuming the current gain of the transistor Q is 25, and that input voltage V_i produces a peak based current of 10mA, calculate the input supply power, output power and efficiency of the amplifier. (L3, S11)



Explain it - Solution 2 (Sheet 2 Q2)



$$I_{BQ} = \frac{V_{CC} - 0.7}{R_B} = 19.3mA$$

Therefore $I_{CQ} = \beta I_B = 25 * 19.3mA = 0.48A$

$$V_{CEQ} = V_{CC} - I_C R_C = 20 - 0.48 * 20 = 10.4V$$

Therefore supply input power is:

$$P_i(dc) = V_{CC} I_{CQ} = 20 * 0.48 = 9.6W$$

Given that input (ac) signal has a peak current of 10mA

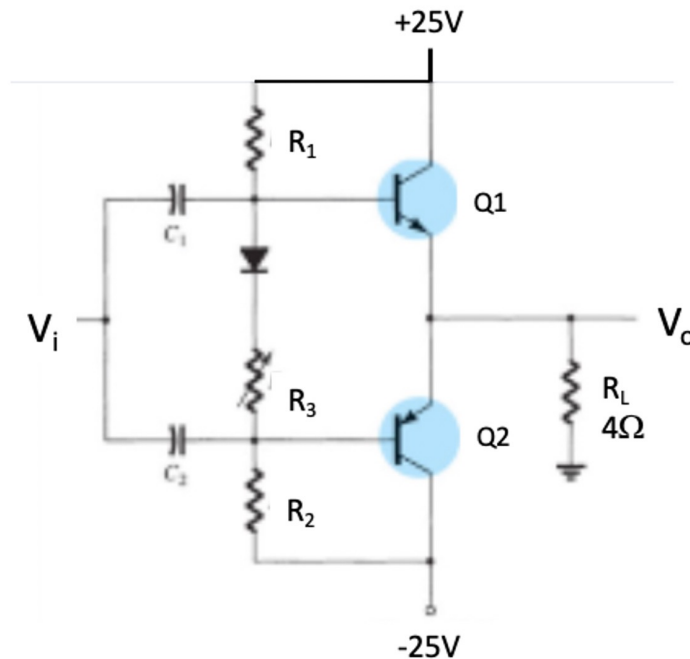
$$I_C(pk) = \beta I_B(pk) = 250mA (pk)$$

$$P_o(ac) = I_C^2(rms) R_C = \frac{I_C^2(pk)}{2} R_C = \frac{0.25^2}{2} * 20 = 0.625W$$

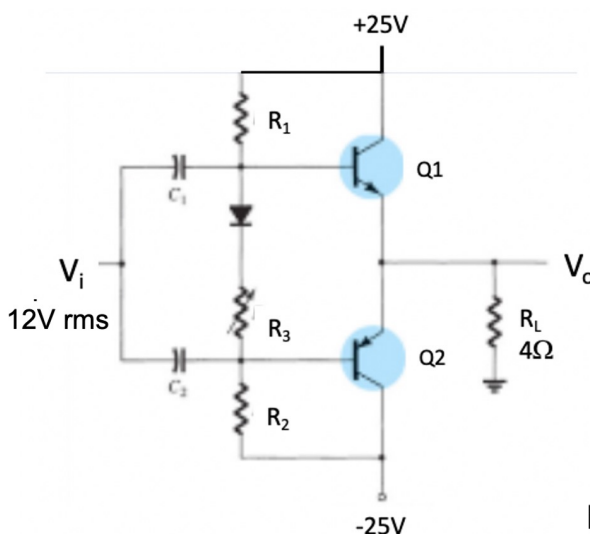
$$\eta = \frac{P_o(ac)}{P_i(dc)} = \frac{0.625}{9.6} * 100\% = 6.5\%$$

Test yourself - Problem 3 (Sheet 2 Q3)

Figure Q3 shows a push-pull amplifier circuit with $V_{CC} = 25V$ and $V_{SS} = -25V$. The input signal (sinewave) is 12V rms, and the gain of the amplifier is 1. Calculate the input power, output power, power efficiency.



Test yourself - Solution 3 (Sheet 2 Q3)



The peak input signal voltage is:

$$V_i(pk) = \sqrt{2} V_i(rms) = \sqrt{2} * 12 = 17V.$$

$$\text{Hence } P_o(ac) = \frac{V_o^2(pk)}{2R_L} = \frac{17^2}{2 \times 4} = 36.125W$$

The peak load current is:

$$I_L(pk) = \frac{V_L(pk)}{R_L} = \frac{17}{4} = 4.25A$$

Therefore the average (dc) current draw from supply rails is:

$$I_{dc} = \frac{2}{\pi} I_L(pk) = 2 \times \frac{4.25}{\pi} = 2.71A$$

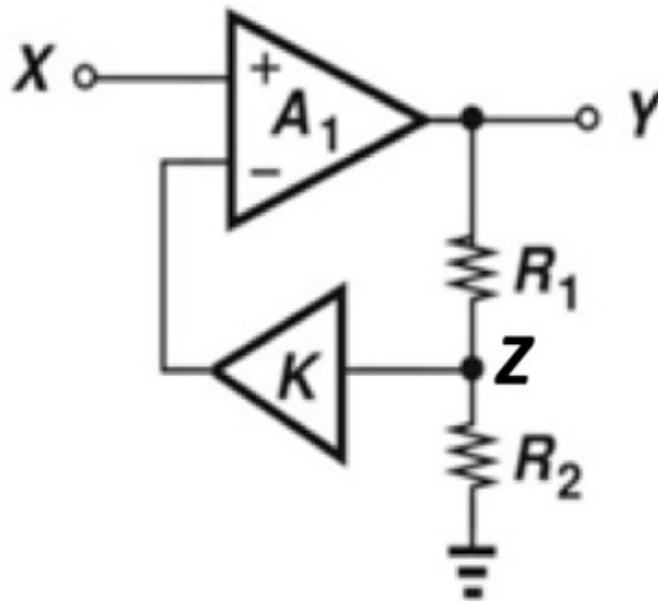
Hence power from supply is

$$P_i(dc) = V_{CC} I_{dc} = 25 \times 2.71 = 67.75W$$

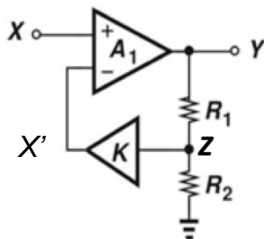
$$\text{The amplifier efficiency is } \eta = \frac{P_o}{P_i} \times 100\% = \frac{36.125}{67.75} \times 100\% = 53.3\%$$

Test yourself - Problem 4 (Sheet 2 Q4)

Derive an equation for the closed-loop gain $G = Y/X$ for the circuit shown below assuming that the open-loop gain of the op-amp is A_1 and the feedback factor is K .



Test yourself - Solution 4 (Sheet 2 Q4)



$$Z = [R_2 / (R_1 + R_2)] * Y = X' / K$$

$$\text{Therefore } X' = [R_2 / (R_1 + R_2)] * K * Y$$

$$Y = A_1 (X - X')$$

$$\text{Therefore } Y = A_1 X - A_1 * K * [R_2 / (R_1 + R_2)] * Y$$

$$Y \{ 1 + A_1 K * [R_2 / (R_1 + R_2)] \} = A_1 X$$

$$\text{Closed-loop gain} = \frac{Y}{X} = \frac{1}{\left(\frac{1}{A_1} + \frac{KR_2}{R_1 + R_2} \right)}$$